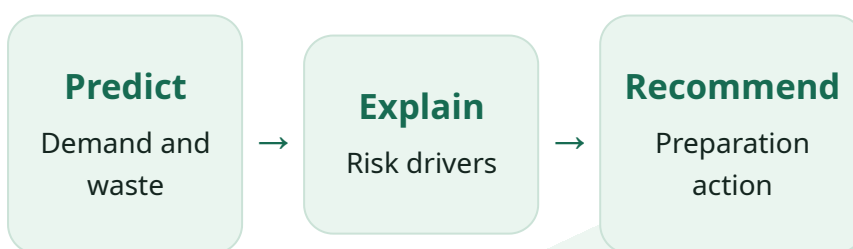


SDG 12 · RESPONSIBLE CONSUMPTION AND PRODUCTION

See tomorrow's demand. Prepare only what is needed.

An AI-powered food-waste prediction and smart-kitchen advisor that converts historical operations data into clear preparation decisions.

WasteLens helps a kitchen predict demand, identify high-risk food items, understand the drivers of waste, and recommend safer preparation quantities before cooking begins.



PROJECT
WasteLens

PREPARED BY
Jatin Panchal

BUILD WINDOW
7-day MVP

SUBMISSION DEADLINE
9 October 2026

PRIMARY PLATFORM
**Responsive Streamlit
dashboard**

WHY

PROBLEM AND PURPOSE

Over-preparation persists because tomorrow's demand is uncertain today

Kitchen teams must decide quantities before they know the actual customer count. When planning relies on intuition or fixed averages, small forecasting errors become unsold meals and avoidable waste.

Demand changes

Attendance, weekday, menu, holidays, events and weather can shift consumption.

Planning stays manual

Past patterns are hard to compare quickly during daily preparation.

Surplus is discovered late

Waste becomes visible only after food has been prepared and service ends.

Learning is lost

Without analysis, the same over-production pattern can repeat.

Core problem statement: Institutional kitchens need a practical way to estimate how much food will be consumed, how much may be wasted, and what quantity should be prepared before the next service period.

Why the problem matters

Operational impact

- Excess ingredients, cooking time and staff effort are spent on food that may not be served.
- Under-preparation is also risky because it can cause shortages and poor service.
- Managers need a balanced recommendation, not a simple instruction to “cook less.”

Sustainability impact

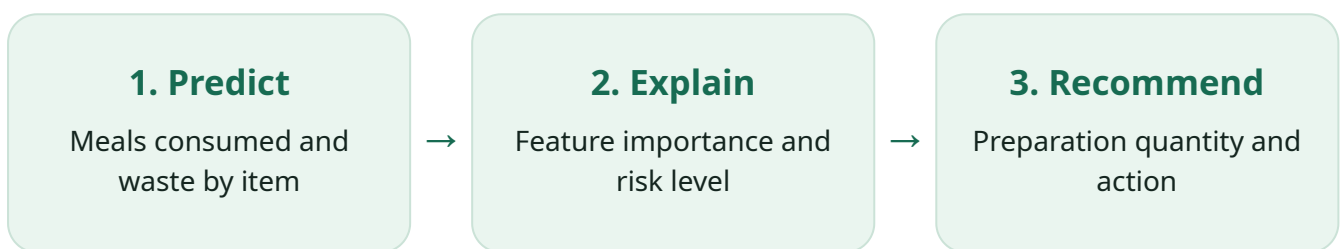
- Preventing avoidable waste supports Sustainable Development Goal 12.
- Item-level visibility helps kitchens act on recurring waste hotspots.
- Transparent predictions can turn sustainability goals into daily operating choices.

WHAT

PROPOSED SOLUTION

A focused Predict-Explain-Recommend workflow is achievable in seven days

WasteLens is a responsive decision-support dashboard. It forecasts demand and food waste, highlights high-risk menu items, explains influential factors, and recommends preparation adjustments.



Demand forecasting

Estimates expected meals using factors such as weekday, prior attendance, meals served, menu, holidays or events, temperature and previous-day consumption.

Food-waste prediction

Uses regression to estimate likely waste overall and by food category, helping users identify the items that need attention first.

Recommendation engine

Applies practical rules or optimization to the predictions and planned quantity, then suggests a safer preparation target and waste-risk level.

What-if simulator

Lets users change expected customers or operating conditions and immediately compare the baseline with the revised preparation scenario.

MVP principle: WasteLens is designed as a decision-support system rather than a fully automated kitchen controller. The manager remains responsible for reviewing and applying each recommendation.

Two practical models and a rules layer keep the MVP credible

The system separates prediction from action: models estimate demand and waste, while a transparent recommendation layer turns those estimates into preparation guidance.

Historical operational data

Date, weekday, meal type, menu item, prepared quantity, served quantity, waste, temperature, holiday or event

Demand model

Expected meals consumed

Waste model

Expected waste by item

Prediction engine → recommendation rules → what-if simulator

Responsive dashboard

Forecast cards, risk charts, explanations, actions and downloadable summary

Proposed technical stack: Python, Pandas, scikit-learn or XGBoost, SHAP, and Streamlit. SHAP or feature importance explains key prediction drivers. The final algorithm should be selected after comparing practical regression models with Mean Absolute Error, Root Mean Squared Error and R^2 .

In the seven-day MVP

- Historical data upload or bundled demonstration dataset
- Cleaning, feature engineering and trend analysis
- Demand and waste predictions
- Item-level risk alerts and recommendations
- What-if simulation and explainability

Deferred beyond the MVP

- Full food redistribution marketplace
- Live inventory, point-of-sale or supplier integration
- Automated purchasing and production control
- Multi-site user accounts and enterprise permissions
- Real-time sensors or hardware deployment

Kitchen decision-makers act; institutions and diners benefit

The MVP is designed for people who plan and supervise repeat meal service in canteens, cafeterias, hostels, offices and similar institutional kitchens.

KM

Kitchen or canteen manager

Needs a daily quantity recommendation, fast visibility into risky items, and evidence for adjusting the plan.

CH

Chef or production planner

Needs actionable item-level quantities that balance shortage risk against over-production.

OP

Operations administrator

Needs trends, model performance and downloadable summaries for review and reporting.

SL

Sustainability lead

Needs measurable waste indicators and recurring hotspot identification to support SDG 12 initiatives.

Secondary beneficiaries

- **Students, employees and busy professionals:** benefit from more consistent meal availability and more responsible food service.
- **Institutional leadership:** gains clearer evidence of operational efficiency and sustainability progress.
- **Project evaluators:** can assess an end-to-end AI/ML workflow with explainable, decision-oriented output.

Typical use moment

A manager reviews tomorrow's expected customers and planned menu, runs the forecast, checks the highest-risk items, adjusts the preparation recommendation, tests a different customer scenario, and downloads the final plan.

Design requirement: The interface should prioritise the next decision - expected demand, highest-risk item, recommended quantity and reason - before secondary analytics. It must remain readable on desktop and mobile widths.

The MVP will be judged by operational usefulness, not accuracy alone

A useful result combines acceptable model performance with recommendations that are understandable, responsive and honest about the available data.

Reduce uncertainty

Forecast expected meals and item-level waste from historical patterns and current conditions.

Improve decisions

Convert predictions into usable preparation quantities and risk alerts.

Build trust

Show why a prediction is high or low so users can review the recommendation.

How success will be judged

Prediction quality

MAE, RMSE and R^2 reported for candidate models.

Decision usefulness

Recommendations are clear, plausible and linked to the forecast.

Waste visibility

Users can identify high-risk items and major prediction drivers.

Scenario speed

What-if changes update without a confusing multi-step workflow.

Interface quality

Core tasks remain readable and usable across desktop and mobile widths.

Demo credibility

Data assumptions, model limits and synthetic fields are disclosed.

Seven-day delivery path

Day 1

Dataset and problem framing

Day 2

Analysis and features

Day 3

Train and compare models

Day 4

Recommendation rules

Day 5

Dashboard build

Day 6

Simulation and explainability

Day 7

Testing, screenshots and presentation

Data and use boundary: If complete operational data is unavailable, the MVP may combine a smaller real dataset with clearly labelled synthetic values. WasteLens remains planning support; a kitchen manager reviews every recommendation before use.